

Chapter - 06

$$5x_1 + 2x_2 + 3x_3 = 7$$

$$2x_1 + 7x_2 + 8x_3 = 5$$

$$3x_1 + 9x_2 + 2x_3 = 6$$

$$\begin{bmatrix} 5 & 2 & 3 \\ 2 & 7 & 8 \\ 3 & 9 & 2 \end{bmatrix}$$

3x3 matrix = $n \times n$ matrix

Here, the no. of variables = no. of equations

We can solve this using Gaussian Elimination, LU Decomposition, Inverse Matrix.

However, if we added another equation:

$$4x_1 + 2x_2 + 5x_3 = 10$$

The matrix would look like this,

$$\begin{bmatrix} 5 & 2 & 3 \\ 2 & 7 & 8 \\ 3 & 9 & 2 \\ 4 & 2 & 5 \end{bmatrix}$$

→ [4x3] matrix

Not a square matrix

[$m \times n$] $m > n$.

We cannot directly solve this using Gaussian Elimination, LU Decomposition, Inverse Matrix since this is not a square matrix.

This system is known as overdetermined system where the number of equations is greater than number of variables. The matrix is not a square matrix.

4 equations
3 variables

$$\begin{cases} 5x_1 + 2x_2 + 3x_3 = 7 \\ 2x_1 + 7x_2 + 8x_3 = 5 \\ 3x_1 + 9x_2 + 2x_3 = 6 \\ 4x_1 + 2x_2 + 5x_3 = 10 \end{cases}$$

$$\begin{bmatrix} 5 & 2 & 3 \\ 2 & 7 & 8 \\ 3 & 9 & 2 \\ 4 & 2 & 5 \end{bmatrix}$$

$m \times n$

$m > n$

Orthogonal

Orthonormality : Orthogonal + Normal

We need to check if it has the following properties :
orthogonality and normality .

$$X = [x_0 \ x_1 \ x_2 \ x_3]$$

[1x4] matrix

Row matrix

$$y = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

[4x1]

Column matrix

$$X = [x_0 \ x_1 \ x_2 \ x_3] \text{ [1x4] matrix}$$

$$X^T = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

→ Transpose matrix where row becomes column and vice versa .

[4x1] matrix

$$y = \begin{bmatrix} 5 & 4 & 3 \\ 1 & 0 & 2 \end{bmatrix} \text{ [2x3] matrix}$$

$$y^T = \begin{bmatrix} 5 & 1 \\ 4 & 0 \\ 3 & 2 \end{bmatrix} \text{ [3x2] matrix}$$

$$O_{(m \times n)} = O^T_{(n \times m)} O_{(m \times n)} = I_{(n \times n)}$$

Square matrix

Orthogonality :

$x^T y = 0$

$\theta = 90^\circ$

$x^T y = |x| |y| \cos 90^\circ$

$x^T y = 0$

↑
not important for exam.

Normality :

$x^T x = 1$
 $y^T y = 1$

Example : Check if the following set is orthonormal.

$S = \left\{ \frac{1}{\sqrt{5}} (2, 1)^T, \frac{1}{\sqrt{5}} (1, -2)^T \right\}$

U matrix

V matrix

Normality : ① $U^T U = I$

$\frac{1}{\sqrt{5}} [2 \ 1] \times \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$= \frac{1}{5} [2 \ 1] \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$= \frac{1}{5} (4+1) = 1$

$$U^T U = I \rightarrow \text{True}$$

$$\textcircled{11} V^T V = I$$

$$= \frac{1}{\sqrt{5}} [1 \ -2] \times \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$= \frac{1}{5} [1 \ -2] \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$= \frac{1}{5} (1+4) = 1.$$

$$V^T V = I \rightarrow \text{True}$$

Orthogonality :

$$U^T V = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \times \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$= \frac{1}{5} (2-2)$$

$$U^T V = 0 \rightarrow \text{True}.$$

Since both normality and orthogonality has been proved so the matrix has orthonormal properties.

Kronecker Delta

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

↳ orthonormal

Least Square

$$Ax = b$$

$$[m \times n]$$

$$A^T A x = A^T b$$

$$[n \times m]$$

$$[m \times n]$$

$$[n \times n]$$

↑
Square matrix

By multiplying with Transpose matrix on both sides we can solve the problem of ~~square matrix~~ overdetermined system and obtain a square matrix.

Example :

$$\begin{aligned} f(-3) &= 0 \\ f(0) &= 0 \\ f(6) &= 2 \end{aligned}$$

$$P_2(x) = a_0 + a_1x + a_2x^2$$

↳ degree is 2

Coefficient matrix
for $P_1(x)$:

$$\begin{bmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{bmatrix}$$

The linear system in
matrix form :

$$\begin{bmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

(3x2) matrix

In order to make this
an overdetermined system
we can make it :

$$P_1(x) = a_0 + a_1x$$

$$P_1(-3) = a_0 - 3a_1 = 0$$

$$P_1(0) = a_0 = 0$$

$$P_1(6) = a_0 + 6a_1 = 2$$

3 equations however
2 variables hence we
took $P_1(x)$ instead of

$$P_2(x)$$

$$A^T A x = A^T b$$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ -3 & 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{bmatrix}$$

$[2 \times 3]$ $[3 \times 2]$

$$= \begin{bmatrix} 3 & 3 \\ 3 & 45 \end{bmatrix} [2 \times 2]$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 \\ -3 & 0 & 6 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 12 \end{bmatrix}$$

$$A^T A x = A^T b$$

$$\begin{bmatrix} 3 & 3 \\ 3 & 45 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 2 \\ 12 \end{bmatrix}$$

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 45 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 12 \end{bmatrix}$$

$$a_0 = 3/7$$

$$a_1 = 5/21$$

$$p_1(x) = \frac{3}{7} + \frac{5}{21}x. \quad [\text{We have}$$

solved this overdetermined system using inverse matrix].

$$\begin{aligned} f(2) &= 3 \\ f(3) &= 5 \\ f(5) &= 12 \\ f(6) &= 15 \end{aligned}$$

overdetermined system

$$p_2(x) = a_0 + a_1x + a_2x^2$$

$$p_2(2) = a_0 + 2a_1 + 4a_2 = 3$$

$$p_2(3) = a_0 + 3a_1 + 9a_2 = 5$$

$$p_2(5) = a_0 + 5a_1 + 25a_2 = 12$$

$$p_2(6) = a_0 + 6a_1 + 36a_2 = 15$$

4 equations, 3 variables.

$$\begin{bmatrix} 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 12 \\ 15 \end{bmatrix}$$

$A \quad x \quad b$

$$A^T A x = A^T b$$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 5 & 6 \\ 4 & 9 & 25 & 36 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \end{bmatrix}$$

$[3 \times 4] \text{ matrix} \quad [4 \times 3] \text{ matrix}$

$$[3 \times 3] \text{ matrix} = \begin{bmatrix} 4 & 16 & 74 \\ 16 & 74 & 376 \\ 74 & 376 & 2018 \end{bmatrix} [3 \times 3] \text{ matrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 5 & 6 \\ 4 & 9 & 25 & 36 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 12 \\ 15 \end{bmatrix}$$

$$= \begin{bmatrix} 35 \\ 171 \\ 879 \end{bmatrix}$$

Augmented Matrix:

$$\begin{bmatrix} 4 & 16 & 74 & 35 \\ 16 & 74 & 376 & 171 \\ 74 & 376 & 2018 & 879 \end{bmatrix}$$

Applying Gaussian Elimination:

$$R_2 \leftarrow R_2 - \left(\frac{16}{4}\right) R_1$$

$$R_3 \leftarrow R_3 - \left(\frac{74}{4}\right) R_1$$

$$\begin{bmatrix} 4 & 16 & 74 & 35 \\ 0 & 10 & 80 & 31 \\ 0 & 80 & 649 & 231.5 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - \left(\frac{80}{10}\right) R_2$$

$$\begin{bmatrix} 4 & 16 & 74 & | & 35 \\ 0 & 10 & 80 & | & 31 \\ 0 & 0 & 9 & | & -16.5 \end{bmatrix}$$

$$9a_2 = -16.5$$

$$10a_1 + 80a_2 = 31$$

$$4a_0 + 16a_1 + 74a_2 = 35$$

QR Decomposition

$$A = QR \quad \text{where } Q = \{q_1, q_2, \dots\}$$

$m \times n$ $\hookrightarrow$ orthonormal property.

Q is an orthonormal set of vectors which we can obtain by Gram - Schmidt process.

Suppose we have a coefficient matrix given,

$$A = \begin{bmatrix} 3 & 1 \\ 6 & 2 \\ 0 & 2 \end{bmatrix}$$

u_1 u_2

Using Gram - schmidt process we will convert these u into q . [$Q = \{q_1, q_2\}$]

Step 1: $P_k = u_k - \sum_{i=1}^{k-1} (u_k^T q_i) q_i$

$\hookrightarrow$ The result obtained will be a scalar dot product.

Step 2: $q_k = \frac{P_k}{\|P_k\|}$

$\hookrightarrow$ Gram - schmidt process.

Steps to solve the problem.

Step - 1 : $k = 1$

$$p_1 = u_1$$

Reason 2. We always start from $i=1$

$$p_1 = u_1 - \sum_{i=1}^0 \rightarrow \text{since } k=1 (k-1, 0)$$

↳ This is not possible because we can't go from 1 to 0.

$$p_1 = u_1 = \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix}$$

$$a_1 = \frac{p_1}{|p_1|}$$

$$|p_1| = \sqrt{(3)^2 + (6)^2 + (0)^2} = \sqrt{45}$$

$$a_1 = \frac{1}{\sqrt{45}} \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix}$$

Step 2 : $k = 2$

$$p_2 = u_2 - \sum_{i=1}^1 (u_2^T a_i) a_i$$

$$p_2 = u_2 - (u_2^T a_1) a_1$$

$$p_2 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \left((1 \ 2 \ 2) \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix} \frac{1}{\sqrt{45}} \right) a_1$$

$$p_2 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \left(15 \cdot \frac{1}{\sqrt{45}} \right) \left(\frac{1}{\sqrt{45}} \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix} \right)$$

$$p_2 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \left(\frac{1}{3} \right) \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix}$$

$$p_2 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$$

$$p_2 = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$$

$$a_2 = \frac{p_2}{|p_2|}$$

$$|p_2| = \sqrt{(0)^2 + (0)^2 + (2)^2} = 2$$

$$a_2 = \frac{1}{2} \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$Q = \begin{pmatrix} a_1 & a_2 \end{pmatrix} = \begin{pmatrix} \frac{3}{\sqrt{45}} & 0 \\ \frac{6}{\sqrt{45}} & 0 \\ 0 & 1 \end{pmatrix}$$

$a_1 \qquad a_2$

- We have used Gram-Schmidt process to convert u into orthonormal set of vectors a_1 and a_2 .

$$Ax = b$$

↳ (m x n)

Previously we have used Transpose matrix to convert A into a square matrix.

$$A^T A x = A^T b$$

$$A = QR$$

$$(QR)^T (QR) x = (QR)^T b$$

$$(Q^T R^T) (QR) x = (Q^T R^T) b$$

$$R^T \boxed{Q^T Q} R x = Q^T R^T b$$

↳ According to property of normality, $Q^T Q = I$.

$$R^T \cdot I \cdot R x = Q^T R^T b$$

$$\boxed{R x = Q^T b} \rightarrow$$

We will use this equation to solve the system by QR Decomposition.

Example : $f(-3) = 0$

$$f(0) = 0$$

$$f(6) = 2$$

In order to make this an overdetermined system, we will consider P_1 .

$$P_1 = a_0 + a_1(x) = f(x)$$

$$P_1(-3) = a_0 - 3a_1 = 0$$

$$P_1(0) = a_0 = 0$$

$$P_1(6) = a_0 + 6a_1 = 2$$

$$\begin{bmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

↳ (3×2) matrix

$$A = \begin{bmatrix} \textcircled{1} & \textcircled{-3} \\ \textcircled{1} & \textcircled{0} \\ \textcircled{1} & \textcircled{6} \end{bmatrix}$$

$u_1 \quad u_2$

We will now convert u_1 and u_2 into their corresponding orthonormal set of vectors v_1 and v_2 using Gram-Schmidt process.

$$P_k = U_k = \sum_{i=1}^{k-1} (U_k^T a_i) a_i$$

Step 1 : $k = 1$

$$P_1 = U_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$a_1 = \frac{P_1}{|P_1|}$$

$$|P_1| = \sqrt{(1)^2 + (1)^2 + (1)^2} = \sqrt{3}$$

$$a_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Step 2 : $k = 2$

$$P_2 = U_2 = (U_2^T a_1) a_1$$

$$P_2 = \begin{pmatrix} -3 \\ 0 \\ 6 \end{pmatrix} = \begin{pmatrix} -3 & 0 & 6 \end{pmatrix} \cdot \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} a_1$$

$$P_2 = \begin{pmatrix} -3 \\ 0 \\ 6 \end{pmatrix} = \begin{pmatrix} \sqrt{3} & \frac{1}{\sqrt{3}} \\ 1/\sqrt{3} & \sqrt{3} \\ 1/\sqrt{3} & \sqrt{3} \end{pmatrix}$$

$$P_2 = \begin{pmatrix} -3 \\ 0 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$P_2 = \begin{pmatrix} -4 \\ -1 \\ 5 \end{pmatrix}$$

$$a_2 = \frac{p_2}{|p_2|}$$

$$|p_2| = \sqrt{(-4)^2 + (-1)^2 + (5)^2} = \sqrt{42}$$

$$a_2 = \frac{1}{\sqrt{42}} \begin{pmatrix} -4 \\ -1 \\ 5 \end{pmatrix}$$

$$Q = (a_1 \ a_2)$$

$$Q = \begin{pmatrix} 1/\sqrt{3} & -4/\sqrt{42} \\ 1/\sqrt{3} & -1/\sqrt{42} \\ 1/\sqrt{3} & 5/\sqrt{42} \end{pmatrix}$$

$\uparrow$ a_1 $\uparrow$ a_2

Now we need to find R before solving the system,

$$R = \begin{bmatrix} u_1^T a_1 & u_2^T a_1 \\ 0 & u_2^T a_2 \end{bmatrix} \rightarrow \text{This will work if } \underline{u_1} \text{ \& } \underline{u_2} \text{ is given.}$$

$$R = \begin{bmatrix} u_1^T a_1 & u_2^T a_1 & u_3^T a_1 \\ 0 & u_2^T a_2 & u_3^T a_2 \\ 0 & 0 & u_3^T a_3 \end{bmatrix}$$

If u_1, u_2 & u_3 is given.

$$U_1^T a_1 = (1 \ 1 \ 1) \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix} = \sqrt{3}$$

$U_2^T a_1 = \sqrt{3} \rightarrow$ We got this value from Step.

$$U_2^T a_2 = (-3 \ 0 \ 6) \begin{pmatrix} -4/\sqrt{42} \\ -1/\sqrt{42} \\ 5/\sqrt{42} \end{pmatrix}$$

$$R = \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{bmatrix}$$

$$R_1 \cdot R^T b = \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -4/\sqrt{42} & -1/\sqrt{42} & 5/\sqrt{42} \end{bmatrix}$$

$\uparrow$ $\uparrow$
 R a

$\begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \rightarrow b$

$$a_0 = 3/7$$

$$a_1 = 5/21$$

Example : $f(1) = 2$

$$f(2) = 3$$

$$f(3) = 6$$

$$f(4) = 4$$

$$P_2(x) = a_0 + a_1x + a_2x^2$$

$$P_2(1) = a_0 + a_1 + a_2 = 2$$

$$P_2(2) = a_0 + 2a_1 + 4a_2 = 3$$

$$P_2(3) = a_0 + 3a_1 + 9a_2 = 6$$

$$P_2(4) = a_0 + 4a_1 + 16a_2 = 4$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 6 \\ 4 \end{bmatrix}$$

↳ (4×3) matrix - overdetermined system.

$$A = \begin{bmatrix} \textcircled{1} & \textcircled{1} & \textcircled{1} \\ \textcircled{1} & \textcircled{2} & \textcircled{4} \\ \textcircled{1} & \textcircled{3} & \textcircled{9} \\ \textcircled{1} & \textcircled{4} & \textcircled{16} \end{bmatrix}$$

$u_1 \quad u_2 \quad u_3$

Step 1 : $k = 1$

$$P_1 = u_1 =$$

$$a_1 = \frac{P_1}{\|P_1\|} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Step 2 : $k = 2$

$$P_2 = U_2 - (U_2^T a_1) a_1$$

$$= \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} - \begin{pmatrix} 1 & 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{pmatrix} \begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{pmatrix}$$

$$P_2 = \begin{pmatrix} -1.5 \\ -0.5 \\ 0.5 \\ 1.5 \end{pmatrix}$$

$$a_2 = \frac{P_2}{|P_2|} = \frac{1}{\sqrt{5}} \begin{pmatrix} -1.5 \\ -0.5 \\ 0.5 \\ 1.5 \end{pmatrix}$$

Step - 3 : $k = 3$

$$P_3 = U_k - \sum_{i=1}^2 (U_k^T a_i) a_i$$

$$P_3 = U_3 - [(U_3^T a_1) a_1 + (U_3^T a_2) a_2]$$

$$U_3^T a_1 = \begin{pmatrix} 1 & 4 & 9 & 16 \end{pmatrix} \begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{pmatrix}$$

$$U_3^T a_1 = 15$$

$$U_3^T a_2 = (1 \ 4 \ 9 \ 16) \begin{pmatrix} -1.5/\sqrt{5} \\ -0.5/\sqrt{5} \\ 0.5/\sqrt{5} \\ 1.5/\sqrt{5} \end{pmatrix}$$

$$= 5\sqrt{5}$$

$$P_3 = \begin{pmatrix} 1 \\ 4 \\ 9 \\ 16 \end{pmatrix} - \left[\frac{1}{15} \begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{pmatrix} + 5\sqrt{5} \begin{pmatrix} -1.5/\sqrt{5} \\ -0.5/\sqrt{5} \\ 0.5/\sqrt{5} \\ 1.5/\sqrt{5} \end{pmatrix} \right]$$

$$P_3 = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

$$a_3 = \frac{P_3}{|P_3|} = \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

$$Q = \left\{ \begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{pmatrix}, \begin{pmatrix} -1.5/\sqrt{5} \\ -0.5/\sqrt{5} \\ 0.5/\sqrt{5} \\ 1.5/\sqrt{5} \end{pmatrix}, \begin{pmatrix} 1.5/\sqrt{5} \\ -0.5/\sqrt{5} \\ -0.5/\sqrt{5} \\ 0.5/\sqrt{5} \end{pmatrix} \right\}$$

$a_1 \qquad a_2 \qquad a_3$

Now for R matrix -

$$R = \begin{bmatrix} U_1^T a_1 & U_2^T a_1 & U_3^T a_1 \\ 0 & U_2^T a_2 & U_3^T a_2 \\ 0 & 0 & U_3^T a_3 \end{bmatrix}$$

$$R = \begin{bmatrix} 2 & 5 & 15 \\ 0 & \sqrt{5} & 5\sqrt{5} \\ 0 & 0 & 2 \end{bmatrix}$$

$$Rx = Q^T b$$

$$\begin{bmatrix} 2 & 5 & 15 \\ 0 & \sqrt{5} & 5\sqrt{5} \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ -1.5/\sqrt{5} & 0.5/\sqrt{5} & 0.5/\sqrt{5} & 1.5/\sqrt{5} \\ 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 6 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 5 & 15 \\ 0 & \sqrt{5} & 5\sqrt{5} \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 5.5 \\ 27.5/10 \\ -3.5 \end{bmatrix}$$

$$2a_2 = -3.5$$

$$a_2 = -1.75$$

$$\sqrt{5}a_1 + 5\sqrt{5}a_2 = \frac{27.5}{10}$$

$$a_1 = \frac{\frac{27.5}{10} - 5\sqrt{5}(-1.75)}{\sqrt{5}}$$

$$a_1 = 11.45$$

$$2a_0 + 5a_1 + 15a_2 = 5.5$$

$$2a_0 = 5.5 - 5(11.45) - 15(-1.75)$$

$$a_0 = -12.75$$

R matrix is upper triangular matrix [Derivation]

We know, $a_k = \frac{p_k}{|p_k|}$

Example,

$$a_1 = \frac{p_1}{|p_1|}$$

$$p_k = |p_k| \cdot a_k$$

$$p_1 = |p_1| a_1 = u_1$$

$$a_2 = \frac{p_2}{|p_2|}$$

$$p_2 = |p_2| a_2 = u_2 - (u_2^T a_1) a_1$$

⋮

$$p_k = |p_k| a_k = u_k - \sum_{i=1}^{k-1} (u_k^T a_i) a_i$$

Rearranging the last term we get,

$$u_k = |p_k| a_k + \sum_{i=1}^{k-1} (u_k^T a_i) a_i \dots \textcircled{i}$$

Taking a_k in both sides we get,

$$u_k \cdot a_k = |p_k| \cdot \underbrace{a_k \cdot a_k}_{\text{orthonormal (1)}} + \sum_{i=1}^{k-1} (u_k^T a_i) \underbrace{a_i \cdot a_k}_{\text{orthogonal (0)}}$$

$$U_k \cdot a_k = |p_k| \dots \textcircled{ii}$$

Combining $\textcircled{i}$ and $\textcircled{ii}$ we get,

$$U_k = (U_k \cdot a_k) \cdot a_k + \sum_{i=1}^{k-1} (U_k^T a_i) a_i$$

$$U_k = \sum_{i=1}^k (U_k^T \cdot a_i) a_i;$$

$$k = 1, 2, \dots, m.$$

We can say that,

$$U_1 = (U_1^T a_1) a_1 = a_1 (U_1^T a_1)$$

$$U_2 = (U_2^T a_1) a_1 + (U_2^T a_2) a_2$$

$$U_2 = (a_1 \ a_2) \begin{pmatrix} U_2^T a_1 \\ U_2^T a_2 \end{pmatrix}$$

$$(U_1 \ U_2) = (a_1 \ a_2) \begin{pmatrix} U_1^T a_1 & U_2^T a_1 \\ 0 & U_2^T a_2 \end{pmatrix}$$

The 3rd term is,

$$U_3 = (U_3^T a_1) a_1 + (U_3^T a_2) a_2 + (U_3^T a_3) a_3$$

$$= (a_1 \ a_2 \ a_3) \begin{pmatrix} U_3^T a_1 \\ U_3^T a_2 \\ U_3^T a_3 \end{pmatrix}$$

So upto 3rd term,

$$(U_1 \ U_2 \ U_3) = (a_1 \ a_2 \ a_3) \begin{pmatrix} U_1^T a_1 & U_2^T a_1 & U_3^T a_1 \\ 0 & U_2^T a_2 & U_3^T a_2 \\ 0 & 0 & U_3^T a_3 \end{pmatrix}$$

So upto nth term,

$$\begin{matrix} (U_1 & U_2 & \dots & U_n) \\ m \times n \end{matrix} = \begin{matrix} (a_1 \ a_2 \ \dots \ a_n) \\ n \times n \end{matrix}$$

$$\begin{pmatrix} U_1^T a_1 & U_2^T a_1 & \dots & U_n^T a_1 \\ 0 & U_2^T a_2 & \dots & U_n^T a_2 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & U_n^T a_n \end{pmatrix}$$

$n \times n$